

Article

Enhancing Early Skin Cancer Detection: A Deep Learning Approach with Multi-Scale Feature Refinement and Fusion

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Abstract

The global incidence of skin cancer is rising, making it an increasingly critical public health issue. Malignant skin tumors such as melanoma originate from pathological alterations in skin cells, and their accurate early-stage segmentation is crucial for quantitative analysis, early diagnosis, and effective treatment. However, achieving precise and efficient segmentation remains a major challenge, as existing methods often struggle to capture complex lesion characteristics. To address this challenge, we propose a novel deep learning framework that integrates the PVT v2 backbone with two key modules: the Spatial-Aware Feature Enhancement (SAFE) module and the Multiscale Dual Cross-attention Fusion (MDCF) module. The SAFE module enhances multi-scale encoder features through a dual-branch architecture, which adaptively extracts offset information to integrate fine-grained shallow details with deep semantic information, thereby bridging the feature gap across network depths. The MDCF module establishes bidirectional cross-attention between decoder and encoder features, followed by multi-scale deformable convolutions that capture lesion boundaries and small fragments across heterogeneous receptive fields, thereby enriching semantic details while suppressing background interference. The proposed model was evaluated on two public benchmark datasets (ISIC 2016 and ISIC 2018), achieving Intersection over Union (IoU) scores of 87.33% and 83.67%, respectively. These results demonstrate superior performance compared to current state-of-the-art methods and indicate that our framework significantly enhances skin lesion image analysis, offering a promising tool for improving early detection of skin cancer.

Keywords: skin cancer; skin lesion segmentation; multi-scale encoder feature



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1. Introduction

In recent years, environmental pollution and increased ultraviolet radiation have significantly contributed to the rising incidence of skin cancers such as melanoma, drawing considerable attention from the global medical community [1,2]. Early diagnosis is crucial for improving treatment outcomes and patient survival rates [2,3]. Traditional diagnostic methods, which primarily rely on clinical observation and tissue biopsy, are often subjective, invasive, and unsuitable for large-scale screening [4,5]. Medical image segmentation technology offers a non-invasive, high-precision alternative for analyzing skin lesions,

providing clinicians with richer and more accurate diagnostic information [6,7] and holding promise for advancing early detection and treatment.

Automatic segmentation of skin lesions plays a vital role in computer-aided diagnosis systems for melanoma [6,7]. Data from 2022 reveal significant geographical disparities in skin cancer incidence, with Australia reporting 37 cases per 100,000 people, China reporting only 0.37 cases, and a global average of 3.2 cases [1,2]. These variations highlight the need for adaptable diagnostic tools that can perform effectively across diverse populations and clinical settings. Public datasets such as ISIC have become critical resources for developing and validating such technologies, enabling researchers to train robust models that generalize across different demographic and geographic contexts [8,9].

Dermoscopy, a non-invasive magnification imaging technique, has become a key tool in melanoma diagnosis [4,5]. Studies indicate that early diagnosis can achieve 10-year survival rates of 84% to 98%, whereas late-stage diagnosis reduces survival rates to only 10% to 15% [3]. However, manual interpretation of dermoscopic images is time-consuming and labor-intensive, underscoring the necessity of automatic segmentation technology to improve diagnostic accuracy and efficiency [6,7]. Automatic skin lesion segmentation aims to perform pixel-level annotation of lesion areas in dermoscopic images, yet this task faces substantial challenges, including variations in skin color and lesion size, texture diversity, and artifacts such as shadows and body hair [6,10]. Early segmentation techniques, such as thresholding and color space analysis, relied on predefined feature extraction and often failed to handle the variability in lesion appearance and interference from external artifacts [10,11].

Against the backdrop of the continuously rising global incidence of skin cancer, achieving precise early segmentation of malignant skin tumors such as melanoma has become critical for enhancing quantitative analysis accuracy, assisting early clinical diagnosis, and optimizing treatment strategies. However, existing segmentation methods often face a trade-off between segmentation accuracy and model practicality. Highly complex models may capture finer features but are difficult to generalize or deploy, limiting their clinical translation. Traditional methods based on threshold segmentation, edge detection, or color-texture feature extraction can achieve preliminary lesion identification [11]. However, they are highly dependent on image quality and struggle to handle interfering factors such as lesion morphological diversity, blurred boundaries, and artifacts (e.g., shadows, hair follicles) [6,10].

Early segmentation techniques, such as thresholding and color space analysis, relied on predefined feature extraction and often failed to handle the variability in lesion appearance and interference from external artifacts [10,11]. Recent advances in deep learning have significantly improved skin lesion segmentation performance. Convolutional neural network (CNN)-based methods, such as U-Net and its variants, have demonstrated strong capability in capturing local features and preserving spatial details through encoder–decoder architectures. Meanwhile, transformer-based approaches, including TransUNet and Swin-Unet, further enhance global context modeling by leveraging self-attention mechanisms, enabling more effective handling of complex lesion structures. Despite these advances, existing methods still face challenges in effectively integrating multi-scale features and preserving fine boundary details, particularly in the presence of noise, low contrast, and irregular lesion shapes. Although deep learning models such as U-Net have significantly improved segmentation performance, they still exhibit limitations in multi-scale feature fusion, long-range contextual modeling, and detail recovery [12,13]. These limitations are particularly evident in cases involving discontinuous boundaries or small lesions, where under-segmentation and mis-segmentation frequently occur [13].

To address these limitations, we propose a novel deep learning segmentation framework that integrates a Pyramid Vision Transformer v2 (PVT v2) backbone with two collaborative modules: Spatial-Aware Feature Enhancement (SAFE) and Multi-scale Dual Cross-attention Fusion (MDCF). This design enhances feature representation and improves decoding reconstruction [14,15]. As shown in Figure 1, we evaluate our method against state-of-the-art approaches on the ISIC 2016 and ISIC 2018 datasets. Experimental results demonstrate significant improvements in both Intersection over Union (IoU) and Dice similarity coefficient (Dice) metrics. The key contributions of our framework are threefold:

First, the PVT v2 encoder leverages its pyramidal hierarchical structure to effectively extract lesion features at multiple scales [15]. This multiscale representation enables the model to adapt to lesions of varying sizes and shapes, from small isolated spots to large irregular regions, thereby improving robustness across diverse clinical scenarios.

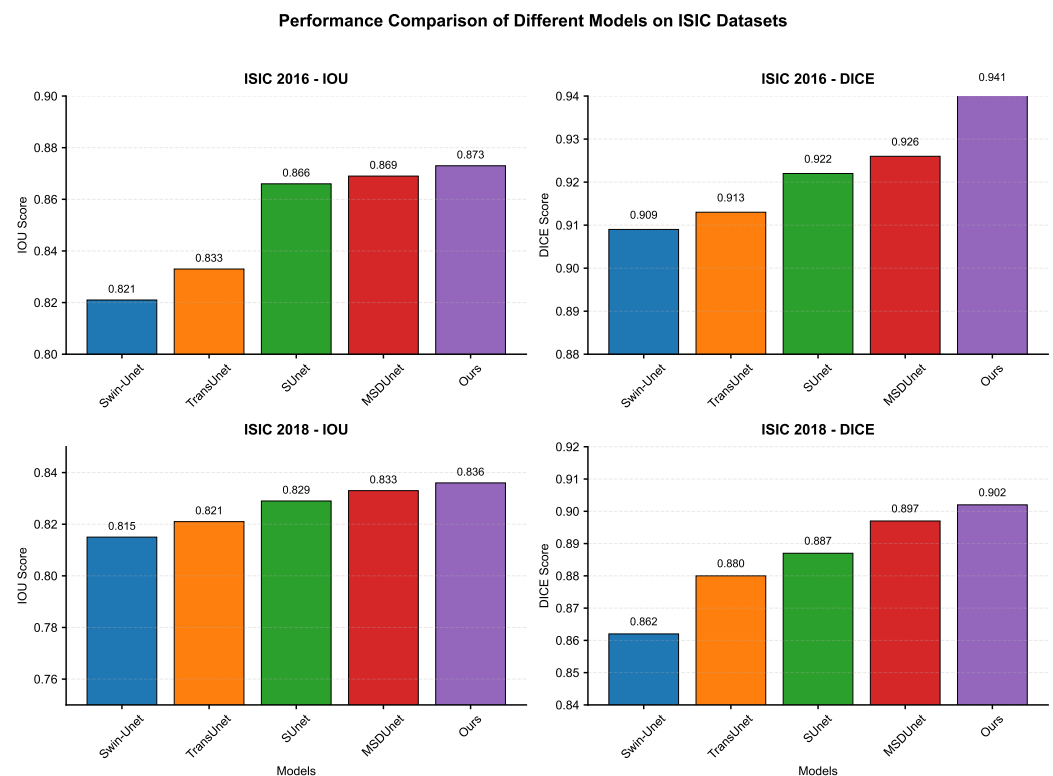


Figure 1. Performance Comparison of Different Models on ISIC Datasets.

Second, the SAFE module refines encoder features at each resolution level through a dual-branch architecture that jointly leverages shallow and deep representations. The base-feature branch preserves fine-grained spatial details from shallow layers, while the offset-compensation branch employs dilated convolutions to predict adaptive offsets that align shallow features with deep semantic information. This mechanism effectively bridges the feature discrepancy across network depths, improving the discriminability and completeness of multi-level feature representation.

Third, the MDCF module establishes bidirectional cross-attention between decoder and SAFE-enhanced encoder features to enable dynamic semantic information transfer via skip connections. It employs a four-branch multi-scale deformable convolution unit to capture lesion boundaries and small fragments at heterogeneous receptive fields. This design significantly improves boundary segmentation accuracy and fine structure recovery while suppressing background responses. In summary, the main contributions of this paper are as follows:

- **Novel Architecture with PVT v2 Backbone:** We propose a new deep learning framework for automatic skin lesion segmentation that incorporates the Pyramid Vision Transformer v2 (PVT v2) as the core encoder. Unlike MSDUnet which uses a hybrid CNN-Transformer (TransXNet) backbone, our choice of a pure transformer-based PVT v2 backbone provides more effective global context modeling through its spatial-reduction attention mechanism, enabling better capture of long-range dependencies in dermoscopic images.
- **Dual-Module Cooperative Mechanism with Distinct Design:** The framework introduces a synergistic dual-module mechanism within the decoder pathway. While MSDUnet employs D2M modules for semantic bridging and MSD Blocks for multi-scale context aggregation, our approach fundamentally differs through two novel components: (1) The Spatial-Aware Feature Enhancement (SAFE) module, which employs a dual-branch architecture to explicitly bridge the feature discrepancy across network depths by adaptively aligning shallow details with deep semantics using offset compensation; and (2) The Multi-scale Dual Cross-attention Fusion (MDCF) module, which establishes bidirectional cross-attention between decoder and encoder features followed by heterogeneous multi-scale deformable convolutions. This design enables more precise boundary delineation and fine structure recovery compared to MSDUnet's densely connected and dilated convolution approaches.
- **Superior Performance on Benchmark Datasets:** We conduct extensive experiments on publicly available datasets, such as ISIC 2016 and ISIC 2018. The results demonstrate that our proposed method achieves state-of-the-art performance.

The remainder of this paper is organized as follows. Section 2 reviews related work on skin lesion segmentation, including both CNN-based and transformer-based approaches. Section 3 presents the proposed segmentation framework in detail, describing the overall architecture as well as the design of the PVT v2-based encoder, the Spatial-Aware Feature Enhancement (SAFE) module, and the Multi-scale Dual Cross-attention Fusion (MDCF) module. Section 4 introduces the experimental setup, including the benchmark datasets, evaluation metrics, and implementation details, followed by comprehensive quantitative and qualitative results, as well as ablation studies and comparisons with state-of-the-art methods. Section 5 discusses the effectiveness, advantages, and limitations of the proposed approach. Finally, Section 6 concludes the paper and outlines potential directions for future research.

2. Related Work

2.1. CNNs for Skin Lesion Segmentation

Convolutional Neural Networks (CNNs) have served as the cornerstone and dominant paradigm in automated skin lesion segmentation, revolutionizing the field through their ability to automatically learn hierarchical features from raw pixel data. The evolution of CNN-based approaches can be largely traced through the adoption and refinement of encoder–decoder architectures [10,12,16,17].

Among these, the U-Net architecture [12] has arguably been the most influential model in medical image segmentation, including skin lesion analysis. Its symmetric encoder–decoder structure, combined with skip connections, effectively addresses the challenge of integrating high-resolution spatial information from the encoder pathway with the high-level semantic information recovered during decoding. Skip connections are particularly critical for preserving fine-grained details along lesion boundaries, which are often irregular and poorly defined. Following U-Net, variants such as SegNet [18]—which utilizes pooling indices for upsampling in the decoder—and other encoder–decoder frameworks have

been widely applied, cementing this architectural paradigm as the standard backbone for the task.

Building upon the encoder–decoder foundation, researchers have introduced various modifications to tackle challenges specific to dermoscopic images. While architectures like U-Net set the standard under full supervision, later work has sought to reduce dependence on costly pixel-level annotations. Weakly supervised methods, for instance, have attracted growing interest. A notable contribution by Anggasta Aji Azhari et al. [19] introduced a two-stage CNN that integrates Grad-CAM for weak localization to refine feature learning. This marks a conceptual shift: rather than using explainability tools only for post-hoc analysis, they are embedded directly into training to improve model confidence and interpretability—bridging high discriminative performance with clinical practicality. Beyond supervision strategies, a key challenge has been enhancing the ability of CNNs to capture long-range contextual information while preserving local detail. While recent efforts have incorporated self-attention mechanisms like Transformers, an emerging direction involves state space models. In this vein, BEFNet [20] employs a Mamba-based encoder to capture global context, while retaining a dedicated CNN branch to preserve local spatial precision—ensuring that critical boundary details are not lost. This reflects a design philosophy of complementary integration, where CNNs play a specialized role alongside global modeling approaches. Furthermore, the computational cost of deep models remains a barrier to clinical deployment. To address efficiency, Chun et al. [21] proposed UCM-Net, which combines MLPs with CNNs in a lightweight architecture requiring under 50 KB parameters and 0.05 GLOPS, yet maintains competitive segmentation accuracy. This makes it a promising benchmark for resource-conscious applications in dermatology.

Another significant challenge in skin lesion segmentation is the severe class imbalance between small lesion regions and extensive background areas. To mitigate this issue, loss functions such as Dice Loss [22] and Focal Loss [23] have been widely adopted as alternatives to standard cross-entropy. These losses explicitly encourage the model to focus on underrepresented lesion pixels, thereby improving segmentation accuracy for the target structures.

2.2. Transformers for Skin Lesion Segmentation

With the rapid advancement of Vision Transformers (ViTs) [24], transformer-based architectures have emerged as powerful alternatives to traditional CNNs in medical image segmentation. Unlike convolutional networks that inherently focus on local spatial relationships, Transformers model long-range dependencies through self-attention, enabling more comprehensive global context understanding—an ability particularly beneficial for skin lesion segmentation, where lesions often present irregular textures and ambiguous boundaries. This advantage has been validated by the Medical Transformer [25], which introduces gated axial-attention to efficiently capture long-range dependencies and has demonstrated strong performance across multiple medical image segmentation tasks.

Hybrid architectures such as TransUNet [14] combine CNN encoders with transformer blocks to incorporate both local feature extraction and global dependency modeling, resulting in improved performance on medical imaging benchmarks. Similarly, hierarchical transformer models like Swin-UNet [26], built upon the shifted-window self-attention mechanism originally proposed in Swin Transformer [27], achieve efficient global modeling while maintaining linear computational complexity with respect to image size.

Pyramid-style transformers further push the performance for dense prediction tasks. The Pyramid Vision Transformer (PVT) and its improved variant PVT v2 [15] utilize spatial-reduction attention and multi-scale feature hierarchies, addressing computational bottlenecks while preserving global contextual cues—making them well-suited for high-

resolution dermoscopic images. Meanwhile, segmentation frameworks such as SegFormer [28] adopt lightweight pyramid Transformers with MLP decoders, demonstrating robust generalization across diverse medical datasets.

Recent studies also incorporate cross-attention modules in the decoder to enhance dynamic feature fusion, improving boundary localization and facilitating better integration of multi-level semantic features. Overall, transformer-based approaches significantly outperform conventional CNNs in capturing long-range structure and resolving fuzzy lesion boundaries, and have thus become a key trend in skin lesion segmentation.

3. Materials and Methods

3.1. Dataset

To comprehensively evaluate the performance of the proposed model, we conducted experiments on three publicly accessible datasets: ISIC-2016 [8], ISIC-2018 [9], and PH2 [18]. The ISIC-2016 dataset contains 900 training and 350 validation images. The ISIC-2018 dataset comprises 2594 images, split into training, validation, and test sets at an 8:1:1 ratio. The PH2 dataset consists of 200 high-quality dermoscopic images, providing a standard benchmark for evaluating the generalization capability of segmentation models.

To analyze the data distribution, we performed a pixel-level statistical analysis across all three datasets. In the ISIC-2016 dataset (2,168,400,305 total pixels), lesion areas account for 17.92% (ratio 1:4.58). In ISIC-2018 (423,886,848 total pixels), the lesion proportion is 21.99% (ratio 1:3.55). For the PH² dataset [29], out of 88,163,323 total pixels, lesion areas occupy 28,435,641 pixels (32.25%), resulting in a lesion-to-background ratio of 1:2.10. The inherent imbalance in this type of dermoscopic image directly motivated our decision to adopt Balanced Accuracy (BACC) and Area Under the Curve (AUC) as evaluation metrics, ensuring a more objective assessment of model performance.

3.2. Architectural Overview

As illustrated in Figure 2, The decoder reconstructs a high-resolution segmentation map from these hierarchical features through a multi-stage refinement process. Starting from the deepest encoder output (Stage 4), feature maps are progressively upsampled and refined at each decoder level. At each resolution, the decoder features are enriched through two complementary mechanisms.

Each lateral connection from the encoder is processed by a SAFE module, which employs a dual-branch architecture to bridge the semantic gap between encoder and decoder features. The base-feature branch preserves fine-grained spatial details from shallow encoder layers, while the offset-compensation branch uses dilated convolutions and global context to predict adaptive spatial offsets, aligning shallow features with deep semantic information before fusion. After receiving SAFE-enhanced encoder features, each decoder stage applies the MDCF module to establish bidirectional cross-attention between encoder and decoder representations. This is followed by a four-branch multi-scale deformable convolution unit that captures lesion boundaries and small fragments at heterogeneous receptive fields, enabling adaptive feature aggregation while suppressing background responses.

A series of convolutional refinement layers and a lightweight prediction head convert the final high-resolution features into a pixel-wise probability map. This design synergistically combines the Transformer's global context modeling with SAFE-guided multi-scale fusion and MDCF-based adaptive aggregation to preserve fine structures while maintaining robust semantic consistency.

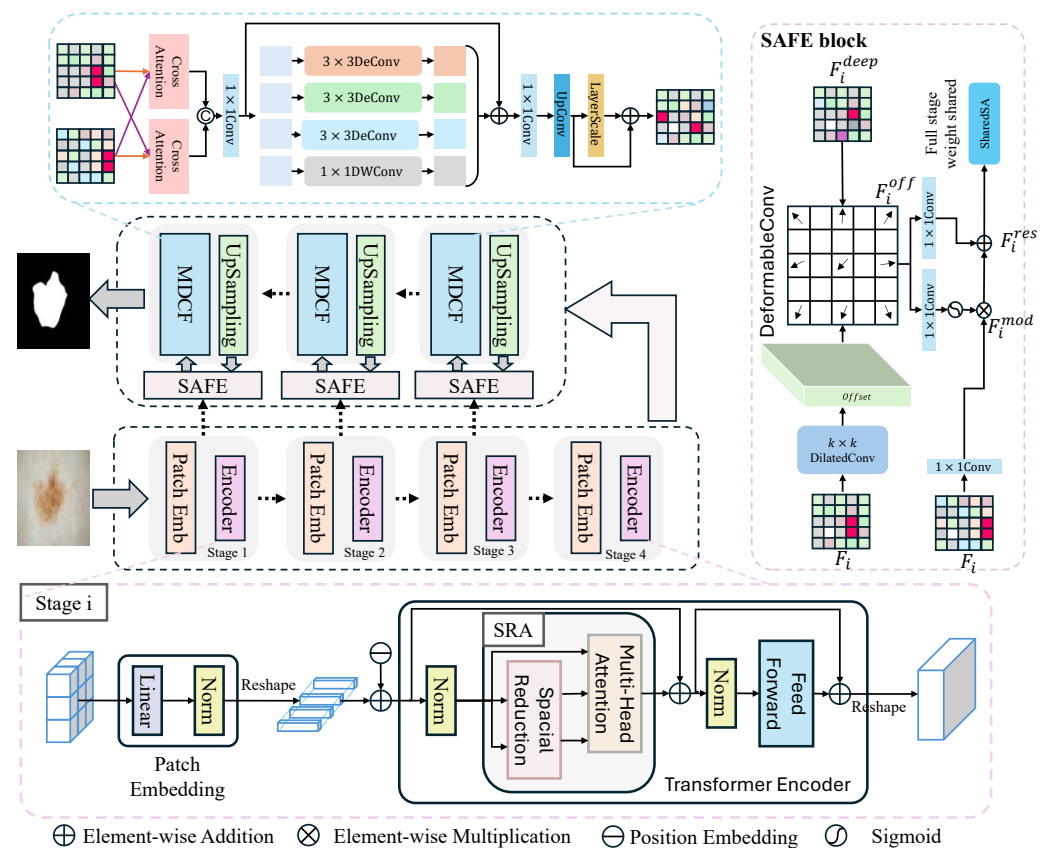


Figure 2. The proposed model architecture.

3.3. Encoder Model

The proposed network adopts a hierarchical Transformer-based encoder to extract multi-scale representations from the input dermoscopic image. As illustrated in Figure 2, the encoder is organized into four stages. Each stage consists of a patch embedding layer followed by a stack of Transformer blocks, and produces a feature map with reduced spatial resolution but increased channel dimension. The resulting feature maps are later used both to drive the decoder and as inputs to the SAFE modules.

At stage i , the input feature map $X_{i-1} \in \mathbb{R}^{H_{i-1} \times W_{i-1} \times C_{i-1}}$ is first processed by a linear projection implemented as a convolutional patch embedding layer. This layer uses a kernel size and stride larger than 1 to merge neighboring pixels into non-overlapping (or slightly overlapping) patches and to down-sample the spatial resolution. A layer normalization is then applied, and the 2-D feature map is reshaped into a sequence of $N_i = H_i W_i$ tokens of dimension C_i . To preserve spatial layout information, we add a learnable 2-D positional embedding to the token sequence before feeding it to the Transformer blocks.

Each Transformer block follows a pre-norm architecture and is equipped with spatial-reduction attention (SRA) to capture long-range dependencies with manageable computational cost. Specifically, the input tokens are first normalized and then fed into an SRA module. In SRA, the query tokens are kept at full resolution, while the key and value tokens are obtained from a spatially down-sampled version of the feature map, enabling global receptive fields with reduced quadratic complexity. The output of the multi-head attention is added to the input via a residual connection. A second normalization layer followed by a position-wise feed-forward network (two linear layers with a non-linearity in between) is then applied, again with a residual connection. Finally, the token sequence is reshaped back to a 2-D feature map to serve as input to the next encoder stage. This process can be expressed by the following formula:

$$T_0^{(i)} = \text{Reshape}(\text{Norm}(\text{Linear}(X^{(i)}))) \quad (1)$$

$$T_{l+\frac{1}{2}}^{(i)} = T_0^{(i)} + \text{MHA}_{\text{SRA}}(\text{Norm}(T_0^{(i)})) \quad (2)$$

$$T_{l+1}^{(i)} = T_{l+\frac{1}{2}}^{(i)} + \text{FFN}(\text{Norm}(T_{l+\frac{1}{2}}^{(i)})) \quad (3)$$

$$Y^{(i)} = \text{Reshape}^{-1}(T_{l+1}^{(i)}) \quad (4)$$

where $X^{(i)}$ denotes the input feature map of the i -th stage, $T_0^{(i)}$ is the token sequence obtained by linearly projecting, normalizing and reshaping $X^{(i)}$, $T_{l+\frac{1}{2}}^{(i)}$ and $T_{l+1}^{(i)}$ are the intermediate features after spatial-reduction multi-head self-attention and the subsequent feed-forward network at the l -th layer, respectively, $Y^{(i)}$ is the output feature map of this stage, $\text{Norm}(\cdot)$ denotes layer normalization, $\text{MHA}_{\text{SRA}}(\cdot)$ denotes multi-head self-attention with spatial reduction, $\text{FFN}(\cdot)$ denotes a position-wise feed-forward network, $\text{Reshape}(\cdot)$ and $\text{Reshape}^{-1}(\cdot)$ convert between 2D feature maps and 1D token sequences.

The encoder outputs from all stages are forwarded to the decoder through SAFE blocks. For a given stage, the corresponding feature map is fed into a SAFE block that exploits both global and local cues. A $k \times k$ dilated convolution is first applied to generate an offset-enriched representation capturing a larger receptive field. In parallel, the global features of the same stage are processed by a weight-shared 1×1 convolution. The two branches are fused and passed through another 1×1 convolution followed by a sigmoid activation to produce a spatial attention map. This attention map is then multiplied element-wise with the original stage feature map, yielding an enhanced representation that emphasizes lesion-related regions while suppressing irrelevant background responses. These SAFE-refined features at multiple scales provide rich contextual information for the subsequent decoder and segmentation head.

3.4. SAFE Model

The SAFE module is designed to refine the encoder features at each resolution level by jointly leveraging shallow-layer and deep-layer representations, while explicitly suppressing background responses. As shown in Figure 2, one SAFE block is attached to the output of each encoder stage, and the resulting enhanced features are forwarded to the decoder.

Given an input feature map from stage (i), denoted as $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$, SAFE first prepares the features and then computes a stage-specific attention map. To reduce computational cost and stabilize training, F_i is optionally passed through a 1×1 convolution and a normalization layer to align the channel dimension. The processed feature map is then fed into two parallel branches: a base-feature branch and an offset-compensation branch. The base-feature branch preserves the shallow representation at stage (i), while the offset-compensation branch predicts offsets to compensate for the discrepancy between the shallow features and the corresponding deep features.

In the base-feature branch, a 1×1 convolution is applied to F_i to obtain a refined shallow representation F_i^{base} . This projection mainly adjusts the channel dimension and preserves the stage-(i) content, so that low-level cues and fine details are maintained in a compact form. A normalization layer and a non-linear activation are used to stabilize training and further refine F_i^{base} , providing a clean base feature for subsequent fusion with deeper layers.

In the offset-compensation branch, SAFE estimates a bias field to align shallow features with their deep counterparts by jointly capturing shallow and deep statistical information. First, a $k \times k$ dilated convolution is applied on F_i to obtain a locally enriched representation

$\mathbf{F}_i^{\text{loc}}$. The dilation rate is chosen according to the spatial scale of the current stage, so that shallow stages focus on fine details while deeper stages capture larger structures. This operation increases the effective receptive field without reducing resolution, thereby aggregating neighboring cues that are important for lesion boundaries and texture. Subsequently, a 1×1 convolution and an activation function project the deep features $\mathbf{F}_i^{\text{deep}}$ into an offset vector, which is then applied to $\mathbf{F}_i^{\text{loc}}$ via a deformable convolution to generate the offset map $\mathbf{F}_i^{\text{off}}$. This map is utilized to modulate the base features, providing a learned bias that compensates for the discrepancy between shallow and deep features.

The outputs of the two branches are then fused to generate a stage-aware representation. Specifically, the offset-compensation branch output $\mathbf{F}_i^{\text{off}}$ is first passed through a non-linear activation function and used to modulate the base features via element-wise multiplication, yielding an intermediate feature $\mathbf{F}_i^{\text{mod}} = \sigma(\mathbf{F}_i^{\text{off}}) \odot \mathbf{F}_i^{\text{base}}$. This modulated feature is then added back to the original offset feature in a residual manner, i.e., $\mathbf{F}_i^{\text{res}} = \mathbf{F}_i^{\text{off}} + \mathbf{F}_i^{\text{mod}}$, and fed into the ShareadSA module to produce the final stage-aware output $\hat{\mathbf{F}}_i = \text{ShareadSA}(\mathbf{F}_i^{\text{res}})$. This fusion strategy enables the learned offsets to gate the shallow base features while preserving offset information, leading to stage-specific refinement for lesion segmentation.

$$\mathbf{F}_i^{\text{loc}} = \text{Conv}_{1 \times 1}(\text{DilatedConv}_{k \times k}(\mathbf{F}_i)) \quad (5)$$

$$\mathbf{F}_i^{\text{off}} = \text{DeformableConv}(\mathbf{F}_i^{\text{loc}}, \mathbf{F}_i^{\text{deep}}) \quad (6)$$

$$\mathbf{F}_i^{\text{base}} = \text{Conv}_{1 \times 1}(\mathbf{F}_i) \quad (7)$$

$$\hat{\mathbf{F}}_i = \text{ShareadSA}(\underbrace{\sigma(\mathbf{F}_i^{\text{off}}) \odot \mathbf{F}_i^{\text{base}}}_{\mathbf{F}_i^{\text{mod}}} + \mathbf{F}_i^{\text{off}}) \quad (8)$$

where $\mathbf{F}_i$ denotes the input feature map at stage i , $\mathbf{F}_i^{\text{off}}$ and $\mathbf{F}_i^{\text{base}}$ denote the features of the offset branch and the base branch, respectively; $\text{DilatedConv}_{k \times k}(\cdot)$, $\text{DeformableConv}(\cdot)$ and $\text{Conv}_{1 \times 1}(\cdot)$ denote a $k \times k$ dilated convolution, deformable convolution and a 1×1 convolution, respectively; $\sigma(\cdot)$ denotes the sigmoid activation function, $\odot$ denotes element-wise multiplication, and $\text{ShareadSA}(\cdot)$ denotes the shared stage-aware attention module.

3.5. Decoder Model

The decoder adopts a cascaded architecture that alternates MDCF blocks with Upsampling blocks, progressively transforming coarse high-level representations into fine-grained, high-resolution predictions. Three MDCF modules are arranged in sequence, each followed by an Upsampling operation (except at the finest scale). At every stage, the MDCF block receives the current decoder feature and its scale-consistent SAFE-enhanced encoder feature, so that spatial resolution is increased while scale-specific encoder information is injected throughout the decoding process.

At the i -th decoding stage, we denote the input decoder feature as $\mathbf{X}_i$ and the corresponding SAFE-refined encoder feature as $\tilde{\mathbf{F}}_i$. The MDCF block first establishes a bidirectional interaction between these two feature streams via a cross-attention mechanism, as illustrated in Figure 2. Specifically, we compute two directional cross-attention responses,

$$\mathbf{Z}_{d \leftarrow e} = \text{CA}(\mathbf{X}_i, \tilde{\mathbf{F}}_i), \quad \mathbf{Z}_{e \leftarrow d} = \text{CA}(\tilde{\mathbf{F}}_i, \mathbf{X}_i), \quad (9)$$

where $\mathbf{Z}_{d \leftarrow e}$ and $\mathbf{Z}_{e \leftarrow d}$ denote the cross-attention feature outputs from the encoder features to the decoder features and from the decoder features to the encoder features, respectively. These two responses are then fused by a 1×1 convolution:

$$\mathbf{X} = \text{Conv}_{1 \times 1}(\text{Concat}(\mathbf{Z}_{d \leftarrow e}, \mathbf{Z}_{e \leftarrow d})). \quad (10)$$

Here, $\text{CA}(\text{feature}_1, \text{feature}_2)$ denotes a cross-attention operator that treats ‘feature_1’ as the query and ‘feature_2’ as the source of keys and values. In a standard implementation,

$$\text{CA}(\mathbf{A}, \mathbf{B}) = \text{softmax}\left(\frac{(W_q \mathbf{A})(W_k \mathbf{B})^\top}{\sqrt{d}}\right) W_v \mathbf{B}, \quad (11)$$

where W_q , W_k , and W_v are learnable projections, and d is the feature dimension. Thus, $\text{CA}(\mathbf{X}_i, \tilde{\mathbf{F}}_i)$ lets each location in the decoder feature selectively aggregate complementary information from the encoder feature, while $\text{CA}(\tilde{\mathbf{F}}_i, \mathbf{X}_i)$ performs the reverse aggregation. This bidirectional design encourages the decoder to focus on encoder regions that are most informative for lesion refinement and to suppress background responses persisting after encoding. The subsequent 1×1 convolution aligns the channel dimension and mixes the information coming from both directions.

As illustrated in Figure 2, the fused feature $\mathbf{X}$ is then refined by a four-branch multi-scale deformable unit. First, $\mathbf{X}$ is evenly split along the channel dimension:

$$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 = \text{Split}(\mathbf{X}), \quad (12)$$

where, if $\mathbf{X}$ has C channels, each sub-feature $\mathbf{x}_j$ contains $C/4$ channels. These four groups are processed in parallel by heterogeneous convolutions:

$$\mathbf{y}_1 = \text{DWConv}_{1 \times 1}(\mathbf{x}_1), \quad (13)$$

$$\mathbf{y}_2 = \text{DeConv}_{3 \times 3}^{(1)}(\mathbf{x}_2), \quad (14)$$

$$\mathbf{y}_3 = \text{DeConv}_{3 \times 3}^{(2)}(\mathbf{x}_3), \quad (15)$$

$$\mathbf{y}_4 = \text{DeConv}_{3 \times 3}^{(3)}(\mathbf{x}_4). \quad (16)$$

The first branch applies a depthwise 1×1 convolution to perform lightweight channel-wise filtering. The remaining three branches, denoted as $\text{DeConv}_{3 \times 3}^{(k)}$ where $k \in \{1, 2, 3\}$, employ deformable convolutions (DCnv2) with progressively increasing dilation rates to achieve heterogeneous receptive fields. Specifically, $\text{DeConv}_{3 \times 3}^{(1)}$, $\text{DeConv}_{3 \times 3}^{(2)}$, and $\text{DeConv}_{3 \times 3}^{(3)}$ use dilation rates of 1, 2, and 3, respectively. This design enables their effective receptive fields to approximate those of standard 3×3 , 5×5 , and 7×7 convolutions, while maintaining the parameter efficiency of a 3×3 kernel. All three deformable branches share a consistent configuration: a kernel size of 3×3 , a stride of 1, and the number of offset groups set to 4. As a result, the three deformable branches possess different effective receptive fields and jointly form a multi-scale detail-enhancement module that is able to capture both thin boundaries and small lesion fragments. The outputs of all branches are concatenated and fused with the original decoder feature via a residual connection:

$$\mathbf{F}^{\text{mdcf}} = \mathbf{X} + \text{Concat}(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4). \quad (17)$$

In this formulation, $\mathbf{F}^{\text{mdcf}}$ denotes the output of the MDCF block; $\mathbf{X}$ is the input decoder feature; $\tilde{\mathbf{F}}_i$ is the SAFE-refined encoder feature; $\text{DWConv}_{1 \times 1}$ is a depthwise 1×1 convolution; $\text{DeConv}_{3 \times 3}^{(i)}$ is the i -th 3×3 deformable convolution branch (not a transposed

convolution); and $\text{Split}(\cdot)$ and $\text{Concat}(\cdot)$ denote channel-wise splitting and concatenation, respectively.

The Upsampling block that follows each MDCF stage increases the spatial resolution of the decoder feature and prepares it for the next, finer decoding level. Concretely, the MDCF output $\mathbf{F}_i^{\text{mdcf}}$ is first passed through a 1×1 convolution to adjust the number of channels and reduce redundancy. The resulting feature map is then upsampled by a factor of 2, either using a learnable deconvolution or an interpolation-plus-convolution scheme, yielding the input feature for the subsequent decoding stage.

3.6. Evaluation Metrics

To evaluate the effectiveness of our method in skin lesion segmentation, the primary evaluation metrics adopted in this paper include the Dice coefficient (also known as Dice Similarity Coefficient, DSC), Intersection over Union (IoU), Sensitivity, Accuracy, Balanced Accuracy (BACC), and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The Dice Coefficient is a statistical metric employed to measure the similarity between two sample sets, which has been extensively applied in medical image segmentation:

$$DC = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (18)$$

The IoU metric quantifies the extent of overlap between the predicted segmentation region and the ground truth annotation:

$$IoU = \frac{TP}{TP + FP + FN} \quad (19)$$

The Accuracy (ACC) metric quantifies the proportion of correctly classified instances, encompassing both positive and negative predictions. The Sensitivity (SEN) assesses the proportion of actual positive cases that are correctly identified.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

$$SEN = \frac{TP}{TP + FN} \quad (21)$$

Balanced Accuracy (BACC) is introduced to provide a more robust evaluation under class imbalance by calculating the arithmetic mean of the recall obtained on each class:

$$BACC = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (22)$$

In addition, the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) is adopted to evaluate the model's discriminative capability across different decision thresholds. It is defined as:

$$AUC = \int_0^1 TPR(FPR) dFPR \quad (23)$$

The performance is evaluated using the true positive rate (TPR, or recall) and the false positive rate (FPR), defined as $TPR = TP / (TP + FN)$ and $FPR = FP / (FP + TN)$. Here, TP (True Positives) and TN (True Negatives) denote, respectively, the counts of positive and negative samples that are correctly classified; conversely, FP (False Positives) and FN (False Negatives) represent the counts of negative samples misclassified as positive and positive samples misclassified as negative, respectively. As these are all counts, the derived

TPR and FPR are dimensionless ratios that provide a reliable assessment of the model's segmentation capability.

4. Results

4.1. Implementation Details

All experiments in this paper are conducted using the PyTorch 1.12.1 framework (with CUDA 11.3 support). The experiments are carried out on a server running Ubuntu 18.04.5 LTS, equipped with two Intel Xeon Gold 6230 CPUs, ten NVIDIA GeForce RTX 3090 GPUs, and a 2 TB solid-state drive.

For all experiments involving Our model, input images are resized to 352×352 pixels before being fed into the network. We employ the AdamW optimizer for model training, with the initial learning rate and weight decay both set to 1×10^{-4} . The network is trained for 200 epochs with a batch size of 8. A step-wise learning rate decay strategy is adopted, where the learning rate is multiplied by a factor of 0.1 every 100 epochs to ensure stable convergence.

To evaluate the intrinsic robustness and feature extraction capability of our proposed framework, no data augmentation techniques (such as random rotation, flipping, or scaling) were employed during the training process. All images were only subjected to standard resizing and normalization before being fed into the network. The loss function is defined as a weighted combination of Binary Cross-Entropy (BCE) loss and Intersection over Union (IoU) loss (formulated here using Dice for structural supervision). The total objective function is defined as follows:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{CE} + \beta \cdot \mathcal{L}_{Dice} \quad (24)$$

where α and β represent the weighting coefficients. To determine the values of the weighting coefficients α and β , we performed a grid search by varying both parameters from 0 to 1 in increments of 0.1. The configuration yielding the best performance was $\alpha = 0.5$ and $\beta = 0.5$, which was subsequently adopted for our experiments. To ensure a fair comparison with other baseline models, we followed their officially recommended training protocols and hyperparameters. The experiments primarily focus on the ISIC-2016 and ISIC-2018 datasets. To provide a more comprehensive evaluation, especially considering the class imbalance, we utilize six metrics: Accuracy (ACC), Intersection over Union (IOU), Dice coefficient (DC), Sensitivity (SEN), Balanced Accuracy (BACC), and Area Under the Curve (AUC).

4.2. Ablation Study

To comprehensively evaluate the contribution of each proposed component, we establish a robust Baseline for performance comparison. The Baseline is defined as the MSDUNet architecture [30], a model characterized by its feature multi-scale and dual-input dynamic enhancement mechanisms. Specifically, MSDUNet employs the TransXNet architecture as its hierarchical encoder backbone to perform feature extraction across four stages (Stages 0–3). Its decoder further integrates D2M (Densely connected to Multi-scale) modules to bridge the semantic gap between encoder and decoder features, along with MSD (Multi-Scale Dilated) Blocks to aggregate multi-scale contextual information.

Using this advanced hybrid CNN-Transformer baseline (achieving an IoU of 86.9% and a DC of 92.6% on the ISIC-2016 dataset) as the starting point, we tested different versions of our model to measure the incremental performance gains provided by our core innovations. We measured performance using six metrics: Accuracy (ACC), Intersection over Union (IOU), Dice coefficient (DC), Sensitivity (SEN), Balanced Accuracy (BACC),

and Area Under the Curve (AUC). The results of the structural ablation experiments are presented in Table 1.

Table 1. Result of an ablation investigation using the proposed method on the ISIC 2016 dataset.

Methods	ACC	IoU	DC	SEN	BACC	AUC
Baseline	0.959	0.869	0.926	0.912	0.958	0.952
PVT v2 backbone	0.963	0.870	0.931	0.915	0.959	0.953
PVT v2 backbone + SAFE	0.966	0.872	0.936	0.918	0.961	0.953
PVT v2 backbone + MDCF	0.968	0.869	0.938	0.920	0.963	0.955
PVT v2 backbone + SAFE + MDCF	0.970	0.873	0.941	0.921	0.964	0.956

The experimental results demonstrate that the Baseline model performs reasonably well in lesion detection, achieving an IOU of 86.9% and a DC of 92.6% on the ISIC 2016 dataset. However, its representation capability for lesion regions is insufficient. When we introduce PVT v2 as the backbone network, the model performance shows noticeable enhancement across all metrics. Specifically, the accuracy increases from 95.9% to 96.3%, the IOU improves from 86.9% to 87.0%, the Dice coefficient rises from 92.6% to 93.1%, and the sensitivity advances from 91.2% to 91.5%. These improvements indicate that the PVT v2 backbone network is more capable of extracting multi-scale features, thereby enhancing the model's representation ability for lesion regions.

Further incorporating the SAFE module on top of the PVT v2 backbone yields additional performance gains. The model achieves an accuracy of 96.6%, an IOU of 87.2%, a Dice coefficient of 93.6%, and a sensitivity of 91.8%. The SAFE module is specifically designed to improve encoder features at each resolution level by jointly leveraging both shallow and deep representations. By explicitly suppressing background responses while enhancing foreground lesion features, the SAFE module addresses one of the key challenges in skin lesion segmentation where background regions often exhibit similar visual characteristics to lesion areas. The integration of multi-level features enables the model to capture both fine-grained details from shallow layers and high-level semantic information from deep layers, resulting in more discriminative feature representations. This synergistic combination of shallow and deep features, coupled with explicit background suppression, leads to more precise localization of lesion boundaries and improved differentiation between lesion and non-lesion regions. As illustrated in Figure 3, the model equipped with SAFE demonstrates superior capability in capturing lesion regions within the images.

The complete model, which combines the PVT v2 backbone, SAFE module, and MDCF module, achieves the best performance across all evaluation metrics. It attains an accuracy of 97.0%, an IOU of 87.3%, a Dice coefficient of 94.1%, and a sensitivity of 92.1%. The MDCF module, through its multi-scale dense connection fusion mechanism, effectively integrates feature information from different hierarchical levels, further enhancing the model's capability to segment lesions with complex morphologies.

The ablation study conclusively demonstrates that each proposed component contributes meaningfully to the overall performance improvement. The progressive introduction of the PVT v2 backbone, SAFE module, and MDCF module results in incremental performance gains, validating the rationality of our design strategy. Moreover, the synergistic effects among these modules collectively drive the advancement of skin lesion segmentation performance, highlighting the effectiveness of our comprehensive architectural design.

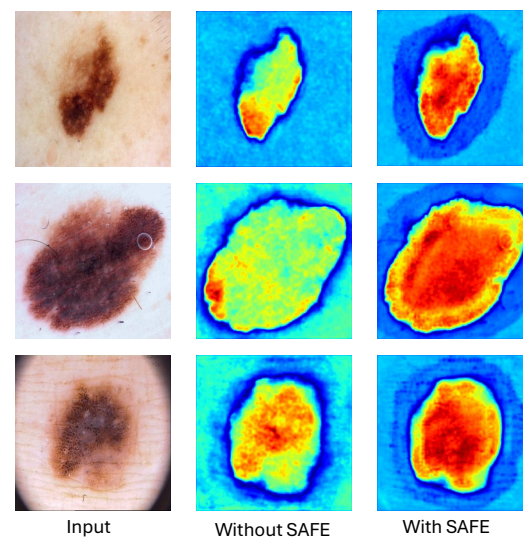


Figure 3. Visualization of decoder features for SAFE model. The deeper the red color, the higher the confidence that the region is a lesion area, while a bluer color indicates lower confidence.

4.3. Comparison with the State-of-the-Art Methods

To demonstrate the effectiveness of our proposed method, we conducted comprehensive comparative experiments against several state-of-the-art segmentation approaches on the ISIC-2016 [8] and ISIC-2018 [9] benchmark datasets. The compared methods include U-Net [12], Swin-Unet [26], TransUNet [14], H2Former [31], SUnet [32], and MSDUnet [30], which represent both classical CNN-based architectures and recent transformer-based models.

Implementation Details for Fair Comparison: All baseline models were re-trained by us under a unified pipeline to ensure a fair comparison. We used the official code for each model with identical data splits and basic augmentations (random flipping/rotation). The key training parameters were set as follows: for Swin-Unet and TransUNet, we used SGD (LR = 0.01, momentum = 0.9, weight decay = 1×10^{-4} , batch size = 24); for H2Former, Adam (LR = 0.001, weight decay = 5×10^{-4} , batch size = 8); for SUnet, Adam (LR = 2×10^{-4} , batch size = 4, 200 epochs with LR decay); for MSDUnet, AdamW (LR = 1×10^{-4} , weight decay = 1×10^{-4} , batch size = 8).

4.3.1. Quantitative Evaluation

Table 2 presents the quantitative comparison results on the ISIC-2016 dataset across six widely-used evaluation metrics: Accuracy (ACC), Intersection over Union (IOU), Dice Coefficient (DC), Sensitivity (SEN), Balanced Accuracy (BACC), and Area Under Curve (AUC). Our proposed method achieves the best performance across all metrics, with ACC of 0.970, IOU of 0.873, DC of 0.941, SEN of 0.921, BACC of 0.964, and AUC of 0.956. Compared to the second-best performing method MSDUnet, our approach demonstrates consistent improvements of 1.1% in ACC, 0.4% in IOU, 1.5% in DC, 0.9% in SEN, 0.6% in BACC, and 0.4% in AUC. When compared to the classical U-Net baseline, our method achieves substantial improvements of 6.8% in ACC, 7.0% in IOU, 5.9% in DC, 3.5% in SEN, 7.0% in BACC, and 5.6% in AUC, which clearly validates the effectiveness of our architectural innovations.

Table 2. A comparative study of different advanced segmentation techniques on the ISIC2016 dataset. **Bold values** indicate the best performance metrics.

Methods	ACC	IoU	DC	SEN	BACC	AUC
U-Net [12]	0.902	0.803	0.882	0.886	0.894	0.900
Swin-Unet [26]	0.923	0.821	0.909	0.899	0.911	0.906
TransUNet [14]	0.932	0.833	0.913	0.901	0.916	0.913
H2Former [31]	0.956	0.852	0.918	0.906	0.931	0.922
SUnet [32]	0.954	0.866	0.922	0.909	0.933	0.927
MSDUnet [30]	0.959	0.869	0.926	0.912	0.958	0.952
Ours	0.970	0.873	0.941	0.921	0.964	0.956

The results on the ISIC-2018 dataset, summarized in Table 3, further confirm the superiority of our proposed method. Our approach consistently achieves the highest performance with ACC of 0.970, IOU of 0.836, DC of 0.902, SEN of 0.928, BACC of 0.948 and AUC of 0.951. The improvements over MSDUnet are 0.8% in ACC, 0.3% in IOU, 0.5% in DC, 0.7% in SEN, 0.3% in BACC and 1.0% in AUC, while the gains over U-Net reach 4.9% in ACC, 3.0% in IOU, 4.4% in DC, 2.9% in SEN, 3.8% in BACC and 4.5% in AUC. These consistent improvements across two different datasets demonstrate the strong generalization capability and robustness of our method.

Table 3. A comparative study of different advanced segmentation techniques on the ISIC2018 dataset. **Bold values** indicate the best performance metrics.

Methods	ACC	IOU	DC	SEN	BACC	AUC
U-Net [12]	0.921	0.806	0.858	0.899	0.910	0.906
Swin-Unet [26]	0.933	0.815	0.862	0.906	0.920	0.913
TransUNet [14]	0.946	0.821	0.880	0.912	0.929	0.922
H2Former [31]	0.959	0.825	0.877	0.910	0.935	0.926
SUnet [32]	0.958	0.829	0.887	0.916	0.937	0.925
MSDUnet [30]	0.962	0.833	0.897	0.921	0.945	0.941
Ours	0.970	0.836	0.902	0.928	0.948	0.951

The quantitative segmentation results summarized in Table 4 further confirm the superiority of our proposed method. All models are trained on the ISIC-2016 and ISIC-2018 datasets respectively, and validated on the PH² dataset as an external test set. Our approach consistently achieves the highest performance across all six metrics under both training settings, with ACC of 0.953, IOU of 0.864, DC of 0.925, SEN of 0.983, BACC of 0.948 and AUC of 0.946 for the model trained on ISIC-2016, and ACC of 0.957, IOU of 0.876, DC of 0.931, SEN of 0.976, BACC of 0.958 and AUC of 0.952 for the model trained on ISIC-2018. For the ISIC-2016 training setting, the improvements over MSDUnet are 0.6% in ACC, 1.3% in IOU, 1.5% in DC, 0.8% in SEN, 0.7% in BACC and 1.1% in AUC, while the gains over U-Net reach 3.8% in ACC, 4.7% in IOU, 3.9% in DC, 9.3% in SEN, 4.3% in BACC and 4.5% in AUC. For the ISIC-2018 training setting, our method outperforms MSDUnet by 0.9% in ACC, 1.4% in IOU, 0.9% in DC, 0.5% in SEN, 1.1% in BACC and 1.1% in AUC, and surpasses U-Net by 4.0% in ACC, 5.5% in IOU, 3.5% in DC, 4.5% in SEN, 4.4% in BACC and 4.2% in AUC. These consistent performance improvements under two different training scenarios, combined with reliable external validation on the PH² dataset, demonstrate the strong generalization capability and robustness of our method.

It is worth noting that transformer-based methods such as Swin-Unet, TransUNet, and H2Former generally outperform the classical U-Net, indicating the advantages of self-attention mechanisms in capturing long-range dependencies for medical image segmentation. However, our method surpasses these transformer-based approaches by effectively

integrating multi-scale feature extraction with enhanced attention mechanisms, enabling more comprehensive representation learning and precise boundary delineation.

Table 4. Quantitative Comparison of Segmentation Performance. Models trained on ISIC-2016 and ISIC-2018 datasets were validated using PH² as external test set. **Bold values** indicate the best performance metrics.

Methods	ISIC-2016						ISIC-2018					
	ACC	IOU	DC	SEN	BACC	AUC	ACC	IOU	DC	SEN	BACC	AUC
U-Net [12]	0.915	0.817	0.886	0.890	0.905	0.901	0.917	0.821	0.896	0.931	0.914	0.910
Swin-Unet [26]	0.921	0.830	0.894	0.932	0.916	0.904	0.928	0.836	0.904	0.942	0.921	0.916
TransUNet [14]	0.925	0.834	0.897	0.944	0.924	0.911	0.932	0.840	0.910	0.951	0.928	0.924
H2Former [31]	0.935	0.841	0.901	0.950	0.929	0.922	0.938	0.844	0.914	0.956	0.932	0.928
SUnet [32]	0.941	0.848	0.905	0.954	0.937	0.931	0.943	0.851	0.920	0.960	0.940	0.936
MSDUnet [30]	0.947	0.851	0.910	0.975	0.941	0.935	0.948	0.862	0.922	0.971	0.947	0.941
Ours	0.953	0.864	0.925	0.983	0.948	0.946	0.957	0.876	0.931	0.976	0.958	0.952

4.3.2. Qualitative Analysis

Beyond quantitative metrics, we provide qualitative visualization results to offer deeper insights into the segmentation performance of different methods. Figure 4 illustrates representative segmentation results on the ISIC-2016 dataset, comparing our method with TransUNet, SUnet, and MSDUnet alongside the ground truth annotations. The visual comparison reveals that our method produces segmentation masks with significantly clearer and more accurate boundaries, particularly for lesions with irregular shapes as shown in the second and third rows. In challenging cases where artifacts or noise are present, such as the example in the third row, our approach demonstrates superior robustness by generating clean segmentation masks that closely match the ground truth, while other methods tend to produce fragmented or incomplete predictions. Furthermore, for lesions with complex morphologies illustrated in the fourth row, our method better preserves the overall shape and structural integrity of the lesion regions, avoiding the over-segmentation or under-segmentation issues observed in competing approaches.

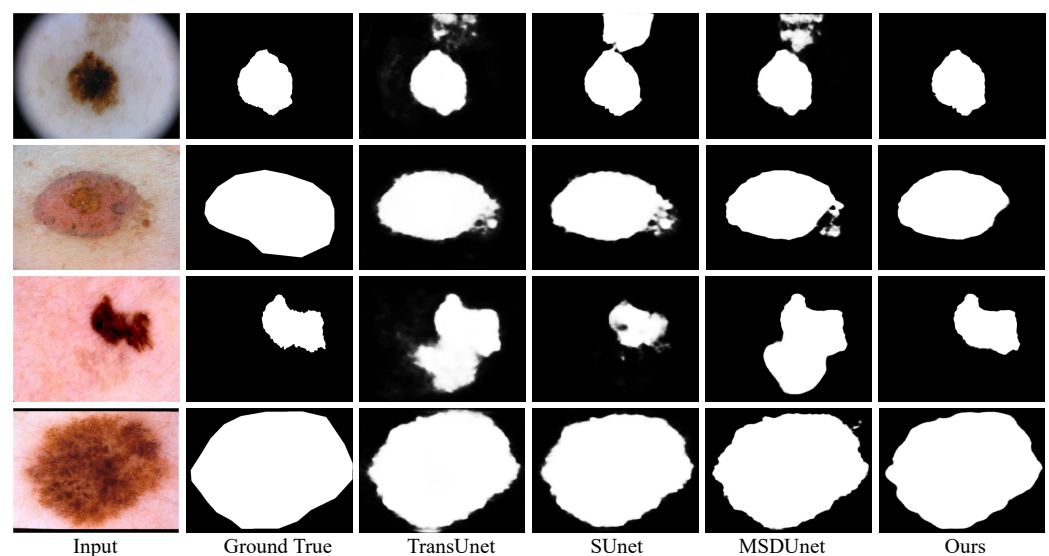


Figure 4. The qualitative analysis on the ISIC-2016 dataset.

Figure 5 presents the qualitative comparison on the ISIC-2018 dataset, which further validates the effectiveness of our proposed method across diverse lesion characteristics and imaging conditions. The second row demonstrates a particularly challenging case

involving a small lesion, where our method successfully achieves precise segmentation while TransUNet exhibits significant under-segmentation, failing to capture the complete lesion extent. For lesions with highly irregular and fuzzy boundaries shown in the first and second rows, our approach produces more coherent and accurate segmentation masks that better align with the ground truth compared to competing methods, which often struggle with boundary ambiguity. Across all four representative cases spanning various lesion types, sizes, and imaging conditions, our method maintains consistent high-quality performance, demonstrating strong generalization capability and robustness to the inherent variability in dermoscopic images.

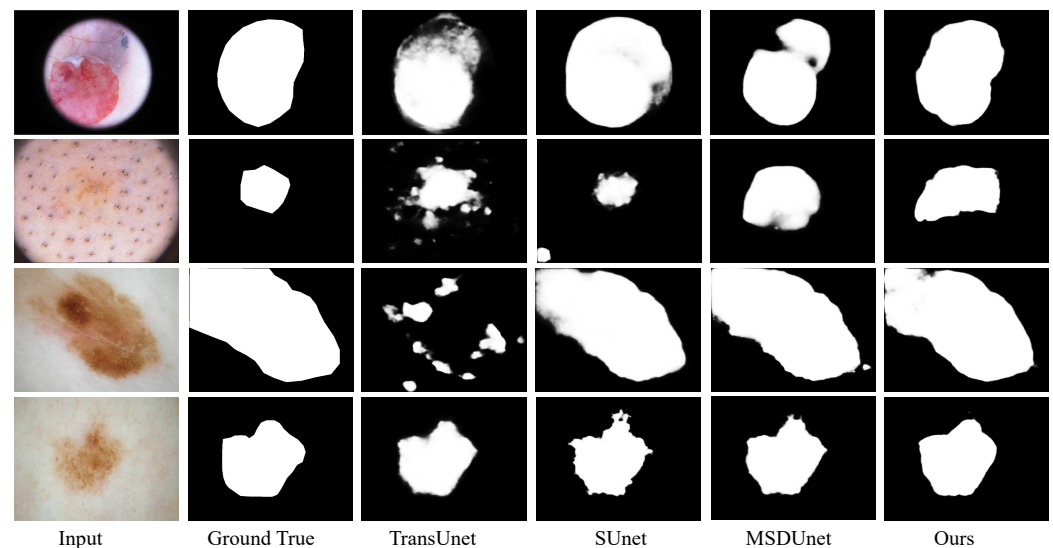


Figure 5. The qualitative analysis on the ISIC-2018 dataset.

Figure 6 presents the qualitative comparison of skin lesion segmentation results, evaluated on models trained with both the ISIC-2016 and ISIC-2018 datasets, which further validates the effectiveness and generalization of our proposed method across diverse dermoscopic image characteristics and dataset distributions. For the ISIC-2016-trained models (first two rows), our method consistently generates segmentation masks that closely align with the ground truth, even for lesions with subtle boundary transitions and inhomogeneous pigmentation. In contrast, competing methods such as TransUNet, SUnet, and MSDUnet exhibit noticeable shape distortion, boundary inaccuracies, and incomplete lesion capture, failing to match the fine-grained details of the ground truth. For the ISIC-2018-trained models (last two rows), our approach maintains superior performance across lesions with varying sizes, shapes, and contrast levels. Notably, in the challenging fourth row case with complex internal texture and irregular boundaries, our method produces a coherent, complete segmentation mask free of the holes and artifacts that plague TransUNet and SUnet, while MSDUnet shows over-smoothing of critical boundary details. Across all representative cases spanning both training datasets, our method consistently outperforms state-of-the-art competitors, delivering more accurate, robust, and visually consistent segmentation results. This confirms that our approach effectively adapts to different dataset distributions and lesion characteristics, demonstrating strong generalization capability and robustness to the inherent variability in dermoscopic images.

The superior performance of our proposed method can be attributed to the synergistic integration of the PVT v2 backbone with the SAFE and MDCF modules within our architecture. The pyramid structure of PVT v2 enables effective multi-scale feature extraction, capturing both fine-grained textural details and global contextual information, which is crucial for handling lesions of varying sizes and appearances. The SAFE module further

refines these multi-scale features by bridging the feature discrepancy across network depths through its dual-branch architecture, where the base-feature branch preserves shallow-layer details while the offset-compensation branch adaptively aligns them with deep semantic information using dilated convolutions and global statistics. The MDCF module then establishes bidirectional cross-attention between decoder and SAFE-enhanced encoder features, facilitating precise localization of lesion boundaries by adaptively emphasizing discriminative features while suppressing irrelevant background information. Its four-branch multi-scale deformable convolution unit captures lesion boundaries and small fragments at heterogeneous receptive fields, which is particularly effective in challenging scenarios with low contrast, ambiguous edges, or irregular morphologies. This comprehensive architectural design enhances the model's ability to handle diverse lesion characteristics—such as variations in size, shape, color, and texture patterns commonly observed in clinical dermoscopic images—thereby achieving superior segmentation accuracy.

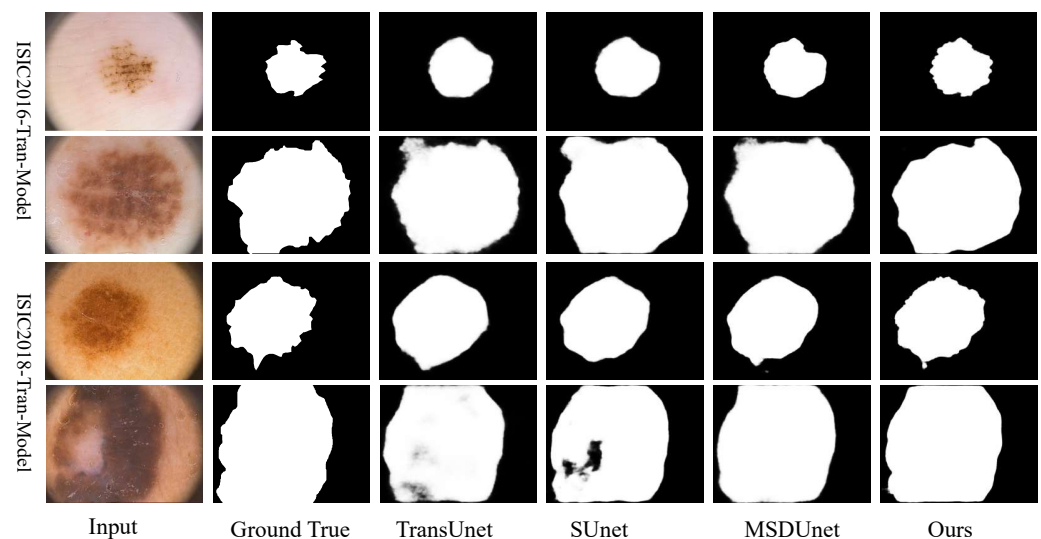


Figure 6. The qualitative analysis on the PH² dataset.

In summary, both quantitative metrics and qualitative visualizations conclusively demonstrate that our proposed method establishes a new state-of-the-art for skin lesion segmentation on the ISIC-2016 and ISIC-2018 benchmark datasets, outperforming existing advanced approaches in terms of segmentation accuracy, boundary precision, and robustness to challenging imaging conditions.

5. Discussion

The experimental results presented in this study demonstrate that our proposed framework achieves state-of-the-art performance on both ISIC 2016 and ISIC 2018 datasets, validating the effectiveness of integrating PVT v2 backbone with SAFE and MDCF modules for skin lesion segmentation. The hierarchical pyramid structure of PVT v2 proves particularly effective for capturing lesion features at different scales, maintaining global contextual awareness. This capability is especially valuable for dermoscopic images where lesions exhibit significant size variation, ranging from small melanomas to large nevi.

When compared specifically to MSDUnet [30], which represents a strong hybrid CNN-Transformer baseline, our method demonstrates clear architectural advantages. While MSDUnet employs TransXNet as its encoder and combines D2M modules with MSD Blocks in the decoder, our approach adopts a pure transformer-based PVT v2 backbone for more effective global context modeling. More importantly, our decoder design introduces two novel modules: the SAFE module explicitly addresses feature discrepancy across

network depths through its dual-branch offset compensation mechanism, and the MDCF module establishes bidirectional cross-attention followed by heterogeneous multi-scale deformable convolutions. These design choices enable our model to achieve superior performance in boundary precision and fine structure recovery, as evidenced by the consistent improvements across all evaluation metrics.

The consistent superior performance on both ISIC 2016 and ISIC 2018 datasets indicates strong generalization capability. Despite differences in image acquisition protocols, lesion types, and annotation quality between these datasets, our method maintains robust performance, suggesting that the learned representations capture fundamental characteristics of skin lesions rather than dataset-specific artifacts. While recent transformer-based approaches demonstrate advantages over classical CNN architectures, our method surpasses them by effectively combining the global modeling capability of transformers with targeted feature enhancement and fusion strategies. The performance gap suggests that architectural innovations beyond simply adopting self-attention mechanisms are necessary for optimal medical image segmentation. The qualitative results reveal that our method excels in challenging scenarios commonly encountered in clinical practice, including lesions with fuzzy boundaries, small lesions that are easily overlooked, and cases with imaging artifacts such as hair or air bubbles. The ability to produce clean, accurate segmentation masks in these difficult cases has direct implications for computer-aided diagnosis systems, potentially reducing false negatives in early melanoma detection and improving diagnostic confidence for dermatologists.

However, several limitations warrant discussion. While our method achieves superior segmentation accuracy, the integration of PVT v2 backbone, SAFE modules at multiple scales, and MDCF blocks with cross-attention mechanisms inevitably increases computational cost compared to lightweight models. Although we did not encounter memory constraints during training, deployment on resource-limited devices may require model compression techniques such as knowledge distillation, pruning, or quantization. Our evaluation is conducted on ISIC challenge datasets which, while widely used as benchmarks, primarily contain images acquired under controlled dermoscopic conditions that may not fully represent the variability encountered in real-world clinical practice. The datasets also exhibit class imbalance, with melanoma cases being relatively rare compared to benign lesions, and inter-annotator variability in ground truth labels may introduce noise affecting both training and evaluation.

The current framework provides deterministic predictions without quantifying uncertainty, which is crucial for clinical decision support. Knowing when the model is uncertain about its predictions can help clinicians identify cases requiring additional scrutiny or expert consultation. Additionally, our model relies solely on image data and does not leverage complementary information such as patient metadata or clinical history that dermatologists consider during diagnosis. While we provide feature visualization to demonstrate that SAFE enhances focus on lesion regions, the overall decision-making process remains largely opaque, and incorporating explainable AI techniques would increase transparency and build trust among medical practitioners. Furthermore, our method is specifically designed for melanoma and common skin lesion segmentation, and its performance on other dermatological conditions has not been assessed. The model also performs single-image segmentation without considering temporal information from follow-up images, which is important for monitoring lesion changes over time.

6. Conclusions

In this paper, we have presented a novel deep learning framework for automatic skin lesion segmentation. The proposed heterogeneous architecture, which synergizes

multi-scale feature enhancement mechanisms, is designed to address key challenges in achieving high segmentation accuracy for complex lesions. By integrating the Pyramid Vision Transformer v2 backbone with Spatial-Aware Feature Enhancement and Multi-scale Dual Cross-attention Fusion modules, our method achieves state-of-the-art performance on the ISIC 2016 and ISIC 2018 benchmark datasets, with IOU scores of 87.33% and 83.67%, respectively.

The PVT v2 encoder leverages its hierarchical pyramid structure and spatial-reduction attention mechanism to extract multi-scale features efficiently, enabling the model to capture both fine-grained textural details and global contextual information essential for handling diverse morphological characteristics of skin lesions. The SAFE module refines encoder features at each resolution level through a dual-branch architecture that preserves shallow-layer spatial details while employing dilated convolutions to predict adaptive offsets that align shallow features with deep semantic representations, effectively bridging the feature discrepancy across network depths. The MDCF module enhances the decoder pathway through bidirectional cross-attention and multi-scale deformable convolutions, establishing dynamic feature fusion that emphasizes relevant lesion characteristics while filtering out redundant background information and capturing lesion boundaries at heterogeneous receptive fields.

Extensive experiments demonstrate that our method achieves superior performance compared to existing state-of-the-art approaches, with both quantitative metrics and qualitative visualizations confirming that our framework excels in challenging scenarios including lesions with irregular shapes, fuzzy boundaries, small sizes, and imaging artifacts. The ablation study validates the incremental contribution of each proposed component, with the complete model achieving the best performance across all evaluation metrics. The clinical implications of our work are significant, as accurate and efficient skin lesion segmentation is a critical step in computer-aided diagnosis systems for melanoma, where early detection can dramatically improve patient survival rates.

Future research directions include developing lightweight variants through model compression techniques to enable deployment on resource-constrained devices, incorporating uncertainty quantification to provide confidence estimates alongside predictions, integrating multi-modal information to enhance diagnostic accuracy, extending the framework to handle temporal analysis for lesion monitoring, exploring weakly supervised learning to reduce annotation requirements, and validating performance on a broader spectrum of dermatological conditions. Our proposed framework represents a significant advancement in automatic skin lesion segmentation, offering a promising tool for improving early detection of skin cancer and demonstrating the potential of deep learning in addressing critical challenges in medical image analysis.

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Conflicts of Interest: Author HuaFu Xu is employed by the Guangxi Zhuang Autonomous Region Bureau of Big Data Development. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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